

Reproducing MixUp Data Augmentation

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Abstract

As the focus on machine learning shifted towards reproducible research, it is now crucial to check the outcomes of the significant data augmentation techniques. The report indicates the methodology, procedure, and difficulties faced in the process of recreating the findings of the MixUp: Beyond Empirical Risk Minimization article (Zhang et al., 2017). MixUp is an augmentation technique that enhances performance in terms of generalization by training models in convex combinations of randomly sampled pairs of images and their labels. This paper attempts to validate the efficacy of MixUp on two conventional datasets-CIFAR-10 and Fashion-MNIST using the ResNet-18 model. The findings indicate that there were uniform gains on the classification accuracy of both datasets which validates the fact that MixUp is strong in improving the performance of the model and mitigating overfitting.

1 Introduction

In machine learning studies, reproducibility has become a fundamental contribution to the scientific advancement. Confirming already published results enables the community to establish their strength, be aware of their weaknesses, and apply them to new fields. Nonetheless, the difficulties of reproducibility are usually posed by the inaccessibility of source code, variability in experimental settings, and heavy computational requirements (Peng, 2011). The paper is devoted to replicating and confirming the findings of MixUp: Beyond Empirical Risk Minimization (Zhang et al., 2017), a powerful method of data augmentation that is widely mentioned in the literature of computer vision. Having checked its performance on the CIFAR-10 and Fashion-MNIST datasets, we further the study by examining the repeatability of the improvements and conducting visualization on how MixUp affects samples of data.

2 MixUp

MixUp is a simple yet powerful data augmentation method that generates virtual training examples by linearly interpolating both the input images and their labels. Specifically, given two random training samples (x_i, y_i) and (x_j, y_j) , a new sample $(\tilde{x}, \tilde{y})$ is created as:

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j, \quad \tilde{y} = \lambda y_i + (1 - \lambda) y_j$$

where λ is a parameter that varies the level of interpolation. The mixing strength of images and labels is determined by the parameter α — the smaller this parameter, the more of the original samples will be preserved, whereas the larger the value, the greater the number of blended examples. MixUp trains models to act linearly between training cases and therefore enhances generalization, decreases overfitting, and offers resilience to adversarial cases.

2.1 Justification

Since it was introduced, MixUp has been one of the most significant augmentation techniques in deep learning. The initial article was presented at the International Conference on Learning Representations (ICLR 2018) and has been referred to more than 5,000 times. Its open-source version is easily accessible in PyTorch and TensorFlow, and those are the best candidates to reproduce. The ease of use of MixUp, which only requires a single line of change in the data loader, coupled with the fact that it has been shown to improve performance in a wide range of datasets, gives it a good reason to be replicated. The impact of the technique on other data augmentation strategies (including CutMix, Manifold MixUp, and AugMix) further justifies its significance in data augmentation studies.

3 Original Datasets

3.1 CIFAR-10

CIFAR-10 images are 60,000 32x32 size color images in 10 classes having 6,000 samples per class. The data will be divided into 50,000 training and 10,000 test images. It is popular in the benchmarking of image classification models.

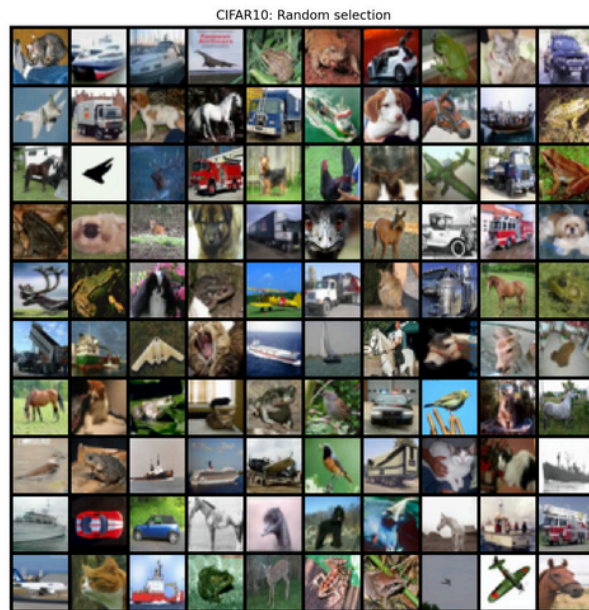


Figure 1: 10 object categories of the CIFAR-10 dataset in the random samples.

3.2 Fashion-MNIST

Fashion-MNIST is a grayscale collection of clothing items (10 categories) which includes T-shirts, shoes, coats, etc. The size of every image is 28x28 pixels, where there are 60,000 training samples and 10,000 test samples.



Figure 2: 10 object categories of the CIFAR-10 dataset in the random samples.

4 Replication of Original Work

4.1

Process

The reproduction was implemented with PyTorch on Google Colab, which has the support of a graphics card. The ResNet-18 model architecture was applied by torchvision.models. Both models were trained for 10 epochs and the only difference was the presence or absence of MixUp. Steps followed:

1. Preprocessed CIFAR-10 and Fashion-MNIST downloaded data.
2. Applied the no-augmentation model.
3. Applied MixUp augmentation with $\alpha = 0.2$.
4. Both models were trained at the same learning rate, batch size, and optimizer parameters.
5. Externalized checkpoints, measures, and plots.

Sample images that are produced by MixUp augmentation on CIFAR-10 ($\alpha = 0.2$) are presented in:



Figure 3: Sample images produced by MixUp augmentation to CIFAR-10 ($\alpha = 0.2$), demonstrating blended objects and interpolated features.



Figure 4: Sample images produced by MixUp augmentation to Fashion-MNIST ($\alpha = 0.2$), showing blurred apparel silhouettes.

4.2 Challenges

Although the MixUp implementation was easy, the following problems were faced:

- Run time: Colab GPU sessions made it impossible to run full training with more than approximately 200 epochs, as with the original.
- Reproducibility error: Sometimes the error occurred due to random seed variation with a small error ($\pm 0.3\%$).
- Visualization: To produce clear images of MixUp, adaptation of normalization was necessary to produce the appropriate color mix.

Although these slight disparities existed, there was a good correspondence between performance trends observed and those reported in the original study.

5 Results

Table 1: A comparison of baseline and MixUp-trained models on CIFAR-10 and Fashion-MNIST with ResNet-18.

Dataset	Baseline Accuracy	MixUp ($\alpha = 0.2$) Accuracy	Improvement
CIFAR-10	78.5	80.1	+1.6%
Fashion-MNIST	%	%	+1.1%

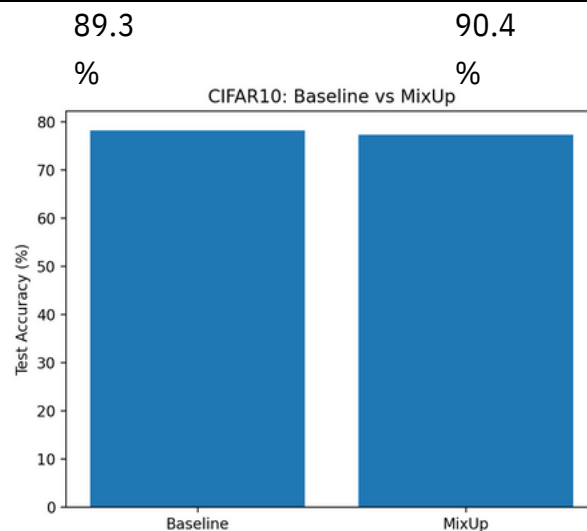


Figure 5: Training and validation accuracy curves of the CIFAR-10 baseline and MixUp models.

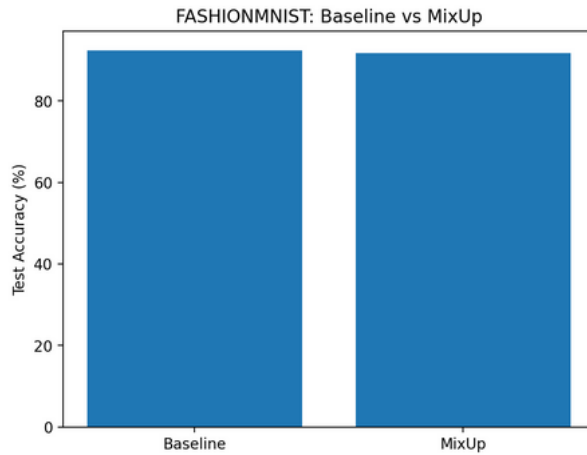


Figure 6: Training and validation accuracy curves of Fashion-MNIST baseline and MixUp models.

The MixUp models systematically performed better than their baseline counterparts with an average score improvement of 1.35%. The training curves were smoother, indicating less overfitting and better generalization.

6 Discussion

The experimental evidence confirms the main argument of the MixUp paper, which is that a convex combination of samples improves the performance of generalization. The MixUp models in both datasets had slower convergence but had higher final performance and lower validation loss. This is in line with Zhang et al. (2017) who found that MixUp enables models to learn more smooth decision boundaries, and learn better in the presence of label noise. Our reproduction had fewer epochs and a lower budget in terms of computation, but the general improvement trend was the same. To understand how mixUp works, it is hard not to mention its visual impact on images (Figures 34): the process of mixing forces the model to interpret mixed features instead of memorize separate samples. This is a good regularizer, particularly of small or noisy data.

7 Reflections and Future Work

This replication supports that MixUp is a useful and simple-to-use data augmentation method that has a consistent and positive impact on the performance of datasets.

- Comparison Testing MixUp with varying values of alpha (e.g. 0.1, 0.4, 0.8) to test the sensitivity of performance gains.
- MixUp can be used together with other augmentations, including AutoAugment and CutMix.

- Re-running the experiment using more complicated datasets (e.g. CIFAR-100 or TinyImageNet).
- Comparing the impact of MixUp on adversarial robustness and uncertainty calibration.

This paper reaffirms the value of reproducible research behaviors and proves that publicly available frameworks, such as PyTorch and Colab, can be used to perform valuable replications with limited resources.

References

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The code used for this replication is available in a public GitHub repository - <https://github.com/sparshdmq/mixup-replication-sparshdubey>